

```
yum install sendmail-cf
service sendmail start
```

- c. VIM or VI Improved is a package that might make your work easier because of the benefit of syntax highlighting. It is very easy to get this package on GNU Linux. On UNIX platforms, we might have to use just the VI editor. (Please check the reference at the end of this article for VIM installations.)
- iv) The following default mail clients can be used:

```
mailx/mail
mutt
OR sendmail -t
```

The above three are the common, basic command line parameters used in UNIX or GNU Linux to send mails. We will be using the last one, *sendmail*, by piping commands to it. This is just to keep things simple and standardised, as we are looking at both UNIX and GNU Linux environments as well as very basic arguments, since all parameters might not be available in all the environments.

The modules normally used in Perl for SMTP are:

```
Net::SMTP
Mail::Sendmail
MIME::Lite (Most widely and commonly used)
```

The procedure

We will avoid importing any Perl modules for actual mailing. Going through these methods, we should be able to learn how a mail is sent out. These are the four types of mails:

1. Clear text email
2. Clear text email with attachment
3. HTML email
4. HTML email with attachment (This is a combination of 2 and 3. Since it is a repetition, we will not actually do this as an example.)

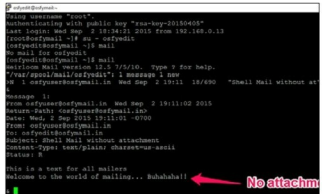
In the examples that follow, do notice the portions that create new lines, in both Shell and Perl. If we miss the right number of new lines, the mail might not produce the desired results.

In Shell, by default, *echo* adds a new line. But in some cases, we need to give one more new line apart from the one Shell already outputs. Perl, by default, does not add a new line; so we have to specifically add the required number of new lines.

In the first case of a clear text email without attachment we might not notice much of a difference. The difference becomes apparent when we have to make more than one MIME addition. Adding files as attachments is one example of going in for MIME additions.

Clear text email without an attachment

On Shell, use the following code:



```
osfyedit@osfyedit:~$ using subsystem "root".
Authenticating with public key "rsa-key-20150405"
Last login: Wed Sep 2 12:24:21 2015 from 192.168.0.13
[osfyedit@osfyedit ~]$ su - osfyedit
[osfyedit@osfyedit ~]$ mail
No mail for osfyedit
[osfyedit@osfyedit ~]$ mail
osfyedit: Mail version 3.2 1/5/10. Type ? for help.
~/var/spool/mail/osfyedit: 1 message 1 new
M 1 osfyuser@osfyedit.in Wed Sep 2 19:11 18/690 "Shell Mail without at
Message 1:
From: osfyuser@osfyedit.in Wed Sep 2 19:11:02 2015
Subject: Shell Mail without attachment
Content-Type: text/plain; charset=us-ascii
Status: R
This is a test for all mailers
Welcome to the world of mailing... Buhahaha!!
[osfyedit@osfyedit ~]$
```

Figure 1: Mail without attachment

```
#!/bin/bash

subject="Shell Mail without attachment";
from="osfyuser"; #can have complete mailing email id
to="osfyedit"; # These are local mail users

bodyFile="mailbody.txt";
cat > $bodyFile <<EOF
This is a text for all mailers
Welcome to the world of mailing... Buhahaha!!
EOF

{
echo "From: $from"
echo "To: $to"
echo "Subject: $subject"
echo -e "Content-Type: text/plain; charset=us-ascii\n"
cat $bodyFile
echo
} | /usr/sbin/sendmail -t
```

- i) As root, create two users, as shown below:
- a. osfyuser

```
useradd osfyuser
```

- b. osfyedit

```
useradd osfyedit
```

- ii) Log in as *osfyuser*. Use the editor of your choice to save the above code as a file. Let us call it *mailer.sh* and make it executable (we're using VIM), as follows:

- a. `chmod u+x mailer.sh`

- iii) Run the executable file, as follows, so that it sends a text mail to the user *osfyedit*:

- a. `./mailer.sh`